GAJADI PRANEESHA

Indian Institute of Technology, Jodhpur

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GitHub

PROJECTS

Ornithopter Mechanism Design and Motion Simulation

Fusion 360 | Motion Analysis | Mechanism Design

- Designed and assembled a 12-component biomimetic ornithopter mechanism in Fusion 360 to replicate flapping wing motion.
- Developed mirrored left- and right-wing assemblies using assembly constraints and component relationships to ensure symmetric motion.
- Applied revolute and rigid joints to accurately simulate linkage motion and mechanical constraints.

Four-Cylinder Internal Combustion Engine Design and Assembly

Fusion 360 | Assembly Modeling | Engineering Drawings

- Modeled 7+ major engine components including the engine block, crankshaft, connecting rods, connecting rod caps, pistons, piston pins, and fasteners using Fusion 360.
- Created a fully constrained four-cylinder engine assembly by applying assembly joints and component relationships to ensure accurate positioning and motion.
- Generated exploded assembly views, detailed engineering drawings, and manufacturing documentation from 3D CAD models.

CAD Design and Manufacturing Drawings of Rigid Flange Coupling

Fusion 360 | Mechanical Assembly | Engineering Drawings

- Designed a rigid flange coupling assembly using Fusion 360 following standard machine design principles.
- Created detailed part models, assembly views, and manufacturing drawings with dimensions.
- Prepared fabrication-ready engineering documentation for all assembly components. .

POSITIONS OF RESPONSIBILITY

Student Guide, Student Wellbeing Committee

- Mentored 11 first-year students by providing academic guidance and campus support.

Assistant Head, Prometeo IIT Jodhpur

- Coordinated publicity and participant outreach for the institute's annual technical festival.

Participant & Mentor, Vandre IIT Jodhpur

- Co-developed an award-winning interactive stall and mentored student teams in the subsequent edition.

EDUCATION

Mechanical Engineering

IITJ

July 2024 – Present

Class XII

Little Flower Junior College

June 2023 96%

Class X

St. Peter's Model School

June 2020 94%

SKILLS

Fusion 360 | SolidWorks | AutoCAD | ANSYS Workbench | Matlab | 3D CAD Modelling | Assembly Design | Engineering Drawing | Motion Simulation | Mechanical Design | Machine Elements | GD&T | GitHub

COURSEWORK

Completed:

Mechanics of Solids | Fluid Mechanics | Manufacturing Processes | Engineering Mechanics | Thermodynamics | Scientific Computations | Machine Learning

Ongoing:

Digital Fabrication | Production Operations & Management | Heat Transfer | Machine Science & Metrology | Kinematics & Dynamics